EventPilot Project Update

README Requirements Completion Status for Team Review

Project: Autonomous Event Planning Team | **Current implementation:** React + FastAPI + Azure OpenAI + LangGraph + MCP + Langfuse + SQLite

Summary: The project is demo-ready. The core manager-led workflow, specialist roles, MCP tools, tracing, review cycle, and persistence are implemented. External production integrations remain intentionally out of scope.

1. Functional Requirements

README requirement	Status	Delivered implementation
Logistics Planning Agent	**Complete**	LangGraph logistics node calls MCP venue search.
Budget & Finance Agent	**Complete**	LangGraph finance node calls MCP budget validation.
Marketing & Promotion Agent	**Complete**	Marketing node calls approval-gated MCP readiness tool.
Schedule Coordination Agent	**Complete**	Schedule node calls MCP conflict checker.
Risk & Operations Agent	**Complete**	Risk node calls MCP operational risk assessment.
Manager-led delegation	**Complete**	Six specialists fan out, converge at the Event Manager, and provide structured context for final synthesis.
Structured delegation and synthesis	**Complete**	Typed LangGraph state carries specialist outputs and final plan.
Plan-Act-Check loop	**Complete**	Manager creates plan; review validates budget/time constraints; graph routes accordingly.
Reflection/refinement	**Complete**	One bounded reflection retry regenerates a plan with review feedback when validation fails.
Conflict/dependency handling	**Complete**	Budget and schedule checks plus manager review prevent invalid final output.
Workflow efficiency	**Complete**	Independent specialists run in parallel; retries are limited to one.

2. Technical and Delivery Requirements

Area	Status	Delivered implementation
Python	**Complete**	Python 3.11 project environment and modular backend.
LangGraph	**Complete**	Compiled state graph with parallel delegation, manager, review, and reflection routing.
LangChain / Azure OpenAI	**Complete**	AzureChatOpenAI structured Pydantic output for final event plan.
Pydantic	**Complete**	Validated event briefs, plan outputs, agendas, budget lines, risks, and activity items.
FastAPI / Uvicorn	**Complete**	Event creation, plan generation, plan retrieval, and budget revision endpoints.
MCP	**Complete**	Internal streamable HTTP server, client, six planning tools, and one venue-catalog resource.
Langfuse	**Complete**	Generation, budget-revision, Azure, and MCP tool observations with input/output, latency, token, and cost details.
Persistence	**Complete**	SQLite stores event briefs and latest plans across backend restarts.
Frontend	**Complete**	React workflow for brief, plan tabs, risk actions, budget revision, approval, and export.
Architecture/workflow documentation	**Complete**	Mermaid architecture diagram, agent roles, and trade-offs in `docs/ARCHITECTURE.md`.

3. Deliberate Scope Boundaries

Item	Reason / next phase
Live Maps, Weather, Calendar, Email	Deferred to avoid additional credentials, external cost, and write-action risk. Current MCP tools are deterministic and read-only for reliable demos.
Production multi-user database	SQLite is appropriate for the course demo. A hosted relational database and authentication are a production enhancement.
CrewAI implementation	Not required by the README. LangGraph was selected as the core orchestration framework and fully demonstrates state machines, delegation, review, and reflection.
Advanced evaluation suite	Langfuse provides observability; future work can add dataset-based quality evaluation and regression tests.

4. Demo Evidence

- Azure plan generation and budget revision traces have been verified in Langfuse.
- MCP tools have been protocol-tested using the MCP Inspector/client.
- Workflow compilation, SQLite storage, and a real Azure workflow run have been validated.